

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix}_{2 \times 2} \times \begin{vmatrix} a & b \\ c & d \end{vmatrix}_{2 \times 2} \times \begin{vmatrix} a & b \\ c & d \end{vmatrix}_{2 \times 2} \times \begin{vmatrix} a & b \\ c & d \end{vmatrix}_{2 \times 2} = ((a \times d) - (b \times c))^4$$